

Alisa Katsionova

Software Engineer · Product Engineering · Systems & Tooling

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SUMMARY

Software Engineer building full-stack products, systems tooling, and data-backed applications across **C++**, **Python**, **TypeScript**, **React**, and **Linux**, with experience shipping production frontend systems, REST APIs, cross-platform tooling, and testable backend services.

EXPERIENCE

Product Engineer, UI/UX Lead & Partner

Jan. 2025 – Present

Innovation 4 Everyone, LLC — SaaS Startup

- Lead frontend architecture and technical implementation across **MyLoops** and **Celo**, shipping customer-facing features with React, Next.js, and SvelteKit from requirements through testing and launch
- Co-developed **Atlas**, a reusable React, Next.js, and Firebase component system used across company products, reducing frontend implementation time by **75%** while improving consistency and maintainability

Software Engineer Intern

Jan. 2025 – May 2025

SimBlocks, LLC

- Refactored core C++ libraries and simulation modules from Windows-specific implementations into **cross-platform CMake** projects supporting reliable Windows and Linux builds
- Optimized CMake build configurations and linker settings, reducing compile times and redundant rebuilds by **60%**
- Built the company's **first Linux development and deployment workflow** using shell scripts, with build-verification tests and environment-validation tooling to identify configuration and dependency failures before deployment

Machine Learning Research Assistant

Jun. 2023 – Aug. 2023

USF Computer Science & Engineering

- Designed and executed GPU benchmarking workflows on **GAIVI**, USF's GPU computing cluster, using Slurm and accelerating GPU-based algorithm evaluation by approximately **40%**
- Automated machine-learning experiments and data analysis using Python, TensorFlow, NumPy, and Pandas

Website & Microsoft Teams Intern

Jan. 2025 – May 2025

UCF Arboretum & Sustainability Initiatives

- Debugged and improved internal Microsoft Power Apps used within Teams, implementing **10+ UX enhancements** and automating approval workflows with Power Automate

PROJECTS

Pokémon Nuzlocke Companion | Python, Lua, PySide6, BizHawk

- Developed a **real-time desktop companion** for Pokémon Platinum that reads live Nintendo DS emulator memory to automatically track party, battle, EV, storage, and Nuzlocke state
- Implemented Generation IV Pokémon decoding directly from **raw emulator memory**, handling encrypted data blocks, block permutation, checksums, stats, IVs, EVs, abilities, held items, and friendship
- Architected modular emulator, memory-decoding, game-state, and UI layers; built RAM diagnostics and automated model/parsing tests with **pytest** and **GitHub Actions**

Full-Stack Banking Application | Python, FastAPI, React, MongoDB, JWT

- Developed a full-stack banking application with **React**, **FastAPI**, and **MongoDB**, supporting authenticated users, account creation, deposits and withdrawals, transaction history, and account-level banking insights
- Structured backend services using **controller**, **service**, **repository**, and **schema layers**, implementing JWT-protected account access and **pytest** coverage for authentication, authorization, and protected routes

Sealbound | Godot, GDScript, Git, Jira

- Led a **five-person cross-functional team** developing a narrative-focused pixel-art RPG in Godot, owning system design, development priorities, and production scope
- Designed and implemented dialogue, relationship, NPC scheduling, and day/night gameplay systems using GDScript

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL, C#, Lua, GDScript, HTML/CSS

Frameworks & Full-Stack: React, Next.js, SvelteKit, FastAPI, Node.js, Firebase, Supabase, MongoDB, PostgreSQL

Systems, Tooling & Cloud: Linux, CMake, Qt/PySide6, Git, GitHub Actions, Vercel, AWS, Slurm

Backend & Engineering: REST APIs, JWT, JSON, pytest, CI/CD, Database Integration, Testing, Technical Documentation, Agile/Scrum

Additional: Godot, TensorFlow, NumPy, Pandas, Accessibility

EDUCATION

University of Central Florida

Bachelor of Science in Computer Science

July 2026